

Jacob Tom

Cambridge, MA • (310) 709-8923 • jtom@college.harvard.edu • [Portfolio: jakettom.github.io/jtom-portfolio](https://jakettom.github.io/jtom-portfolio)

Education

Harvard University

Cambridge, MA

Bachelor of Science, Electrical Engineering. GPA 4.0

June 2027

Relevant Courses: Computer Hardware & Architecture; Control Theory; Circuits and Devices; Signals; Wave Physics; Electronic/Photonic Devices

Honors: John Harvard Scholarship (top ~5% of class), Detur Book Prize (top ~10% of class)

Work Experience

Diffraction

Somerville, MA

Optical / Electrical Engineering Intern

May 2026 – Present

- Designed 2 scene-generation systems using Arduino and Raspberry Pi, including custom optical-table mounts modeled in Fusion 360
- Designed 3 light-source PCBs and 1 power-control PCB for camera-testing optical setups
- Designing and maintaining multiple camera-testing setups incorporating DMDs and SLMs on optical tables
- Programmed an Arduino-based camera sensor for SLM scene generation and designed its optical configuration

Harvard University

Allston, MA

Computing Hardware Teaching Assistant (Lab Assistant)

September 2026 – Present

- Teaching students in System Verilog (HDL) coding for computer architecture design lab (ALUs, MIPS architecture)

Harvard Microrobotics Lab

Allston, MA

Research Assistant

February 2026 – Present

- Currently designing KiCAD PCB for VHF and GPS tracking (embedded sensing, power, RF hardware)

Harvard Recreation

Cambridge, MA

Rock Wall Manager

October 2024 - Present

- Led transition of Harvard Climbing Wall from independent organization to Harvard Recreation operation

Relevant Projects

Senior Capstone Design Thesis: Loudspeaker Project

March 2026 – Present

- Collaborating with Harvard faculty and industry professionals (Bose, Meta Reality Labs) to solve the transportation & storage issues of large-scale speakers

Interferometry-based Optical Profilometer

August - December 2025

- Designed and aligned optical profilometer using lenses, laser source, and optical table instrumentation
- Measured surface profiles with ~100 nm-scale resolution
- Developed a Python digital signal processing pipeline to extract surface profile from optical interference patterns

Multi-Cycle MIPS Processor (on FPGA Board)

August - December 2025

- Implemented multi-cycle MIPS processor on an FPGA (datapath, control logic, ALU)
- Created Python assembler for R-, I-, and J-type instructions
- Developed logic-gate-based ALU supporting arithmetic, shifts, comparisons, error flags

Mars Rover (Harvard University Robotics Club)

August 2024 – January 2026

- Designed a power distribution board in KiCad, included buck converters, fuses, WAGO connectors

Skills

Hardware: PCB Design (KiCad), Analog Circuit Design, Lab Equipment, LTspice, SolidWorks, Soldering

Embedded: Arduino, ESP32, Raspberry Pi

Software: Python, SystemVerilog, MATLAB, C/C++, Git/GitHub